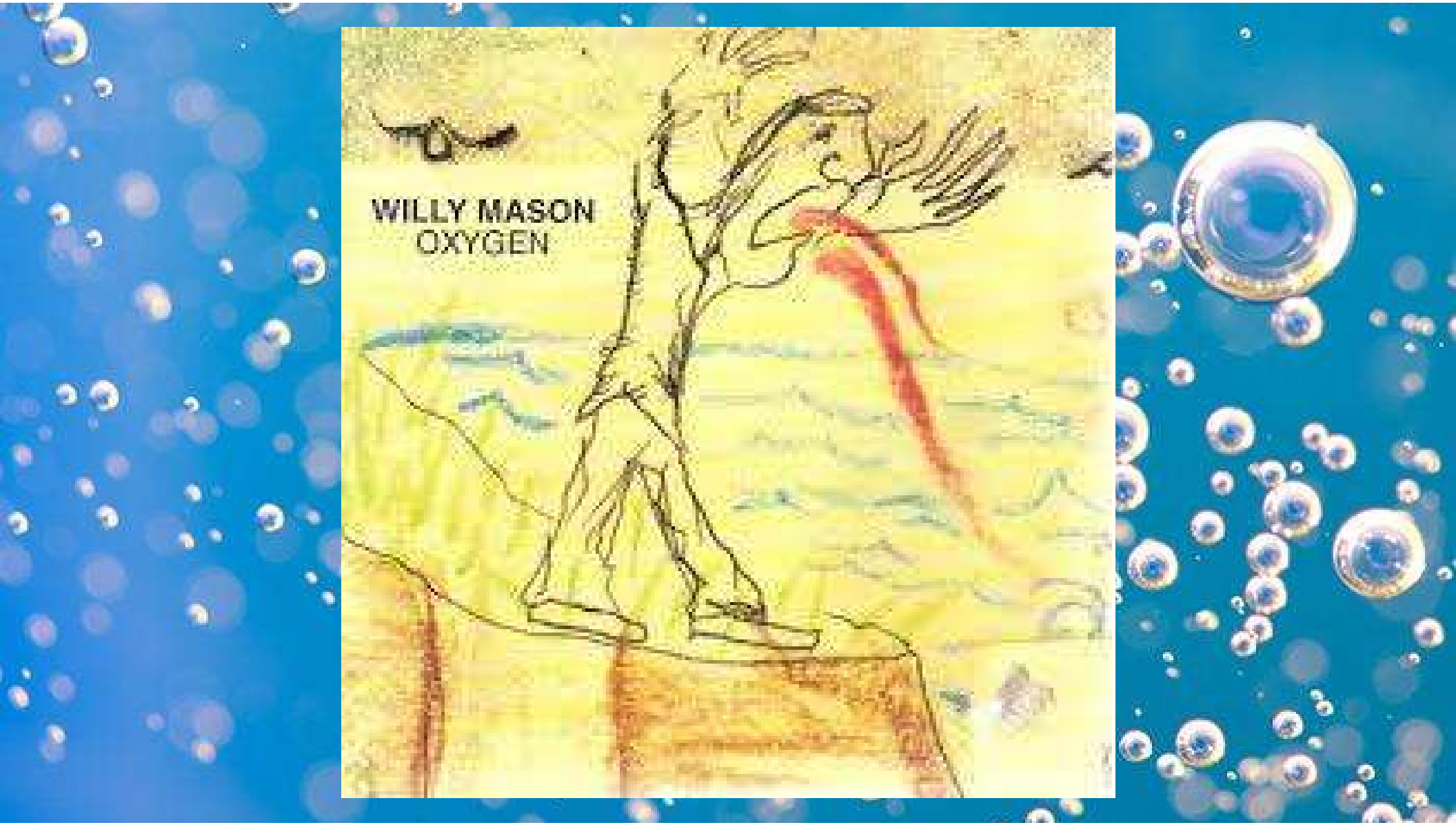
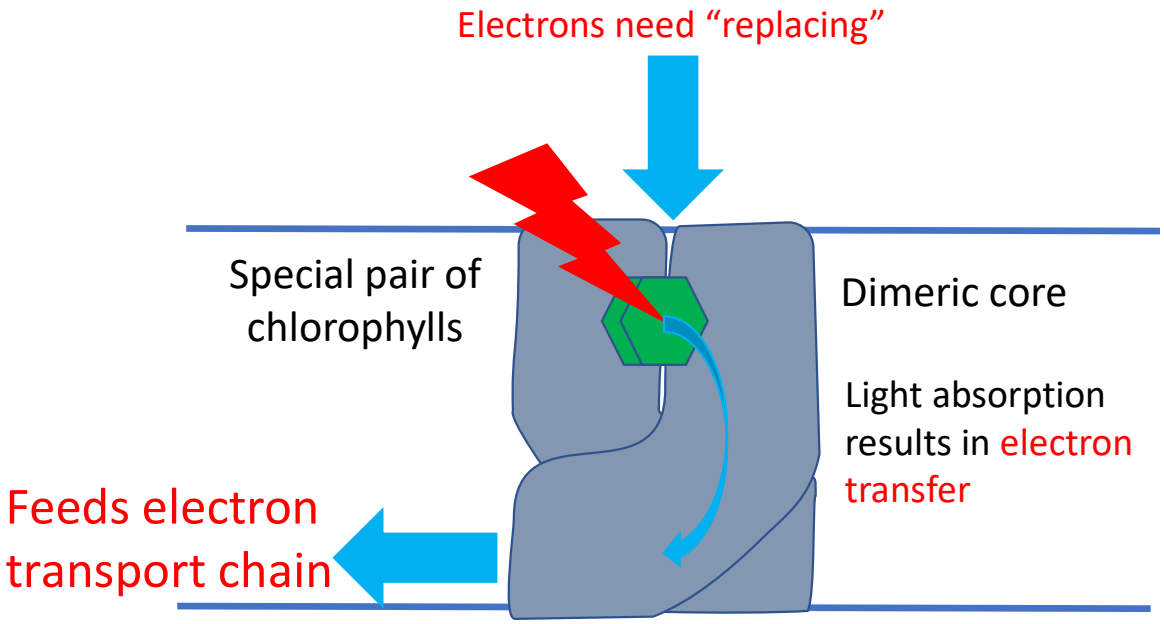
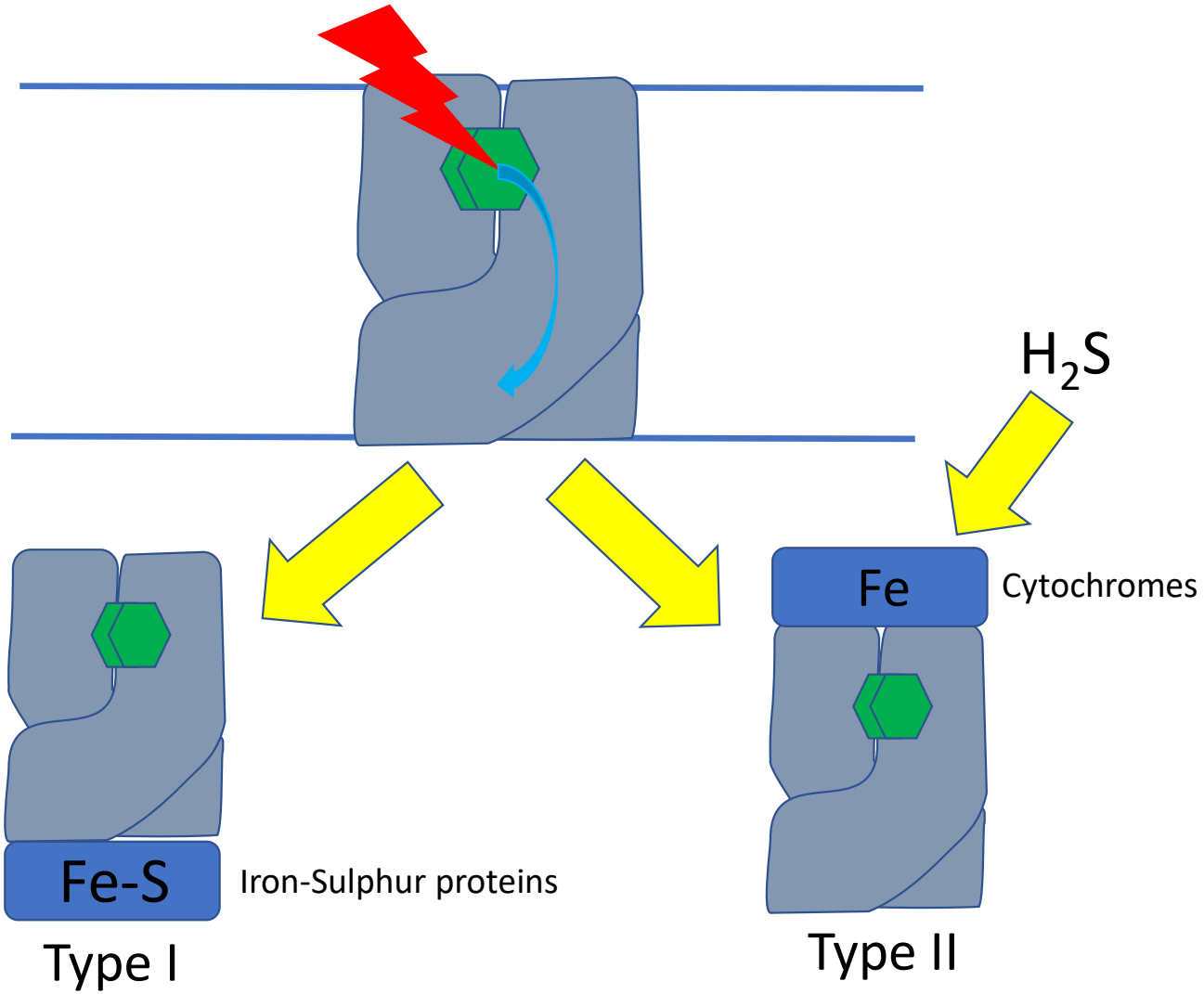

WILLY MASON
OXYGEN

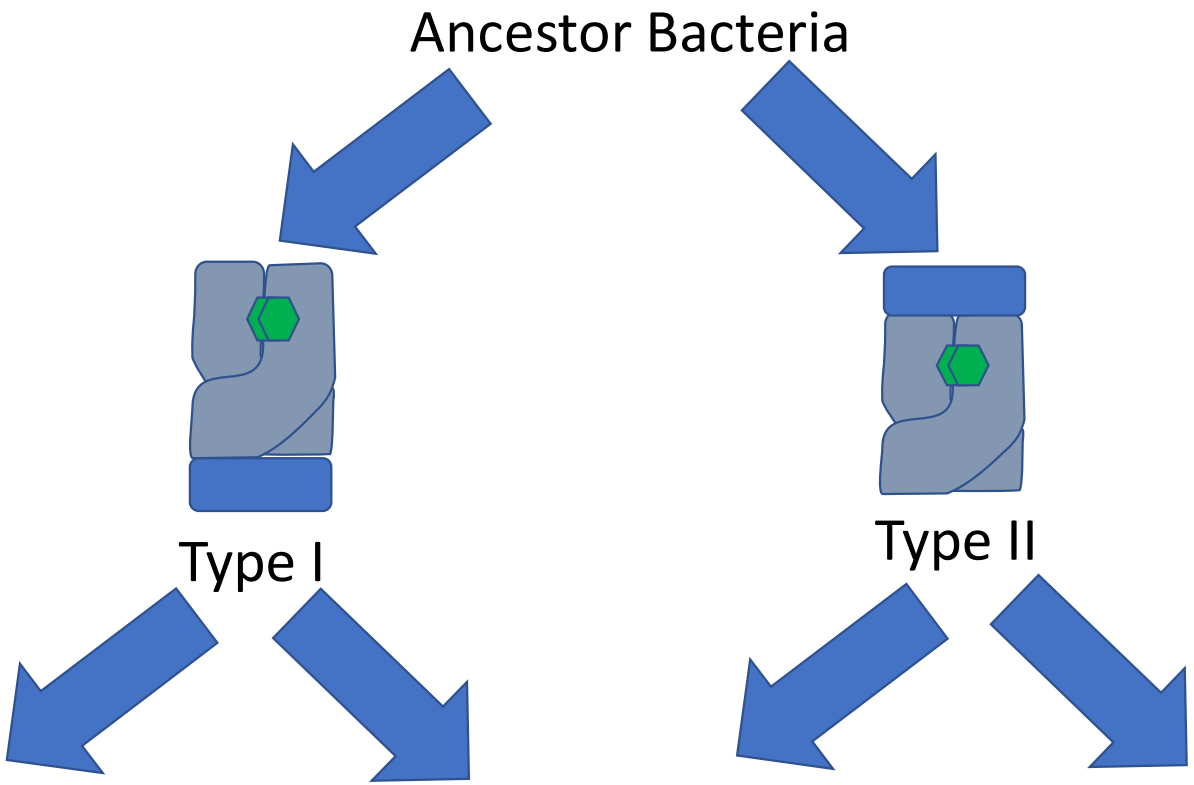
Reaction centres



Reaction centres



Evolution of reaction centres



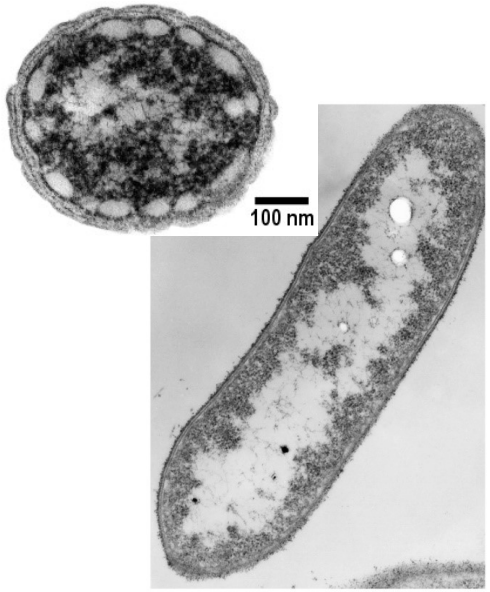
Type I and Type II centres evolved
in different lineages

Type I reaction centres

Found in:

Green sulphur bacteria

Heliobacteria

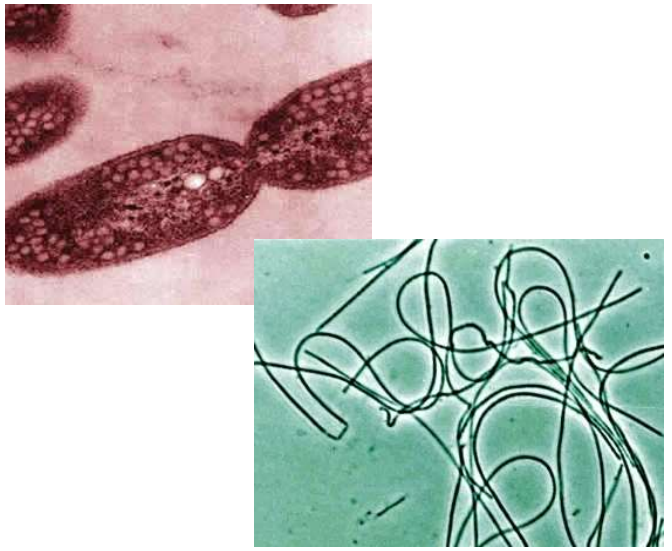


Type II reaction centres

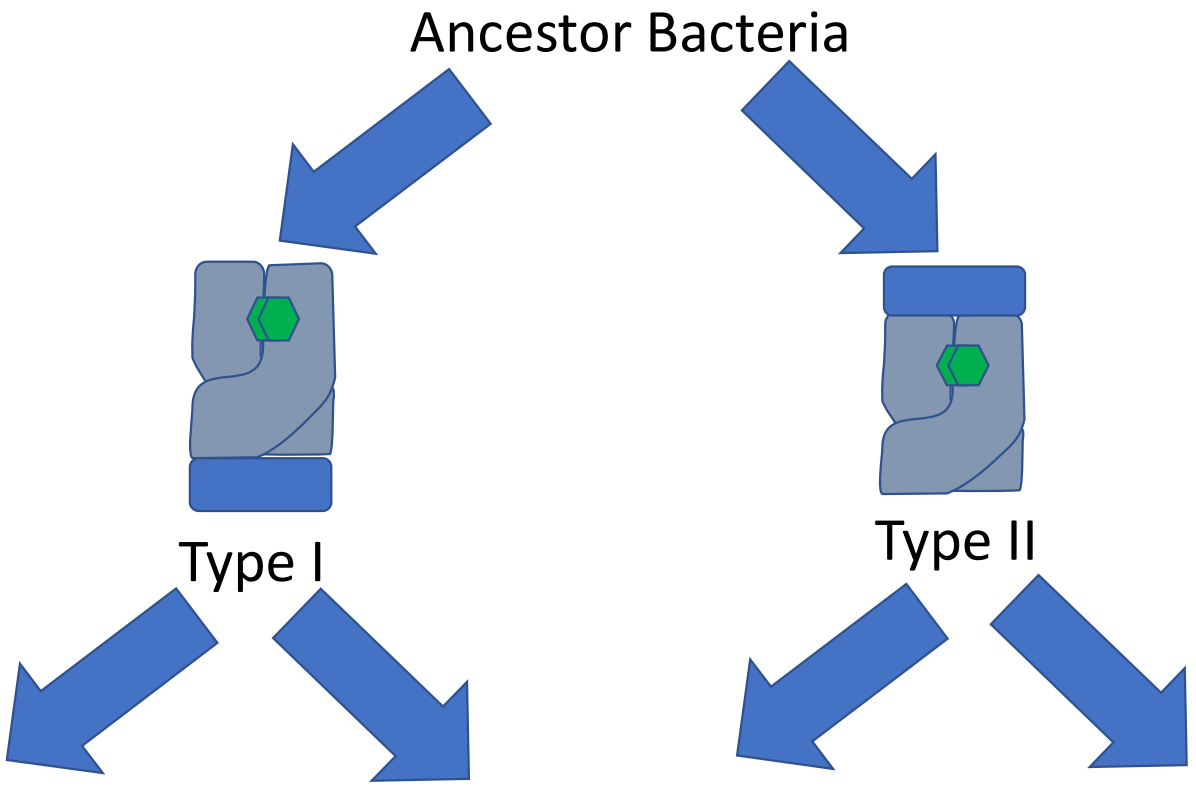
Found in:

Purple sulphur bacteria

Green non-sulphur bacteria



Evolution of reaction centres



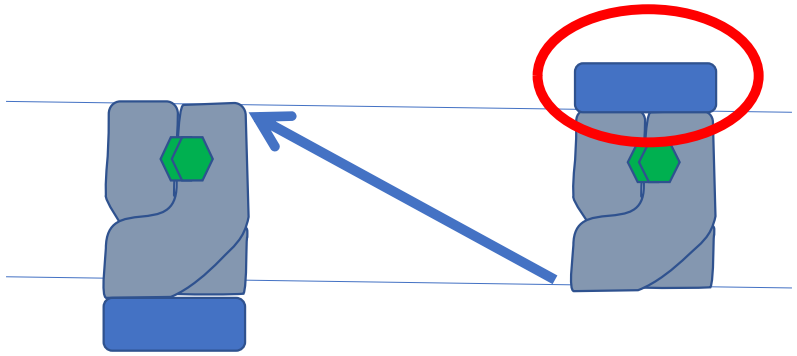
Type I and Type II centres evolved
in different lineages

Except…

Cyanobacteria

"Blue-Green algae"

Contain Type I and Type II reaction centres



Oxygen Evolving Complex

How?

"Lateral gene transfer"

$$2H_2O + \text{Light} \rightarrow 4e^- + 4\ H^+ + O_2$$

Cyanobacterial Fossils

Stromatolites



Can be formed by growth of microbial mats
Early fossil evidence of life

https://en.wikipedia.org/wiki/Stromatolites



https://learningonscreen.ac.uk/ondemand/index.php/clip/101164

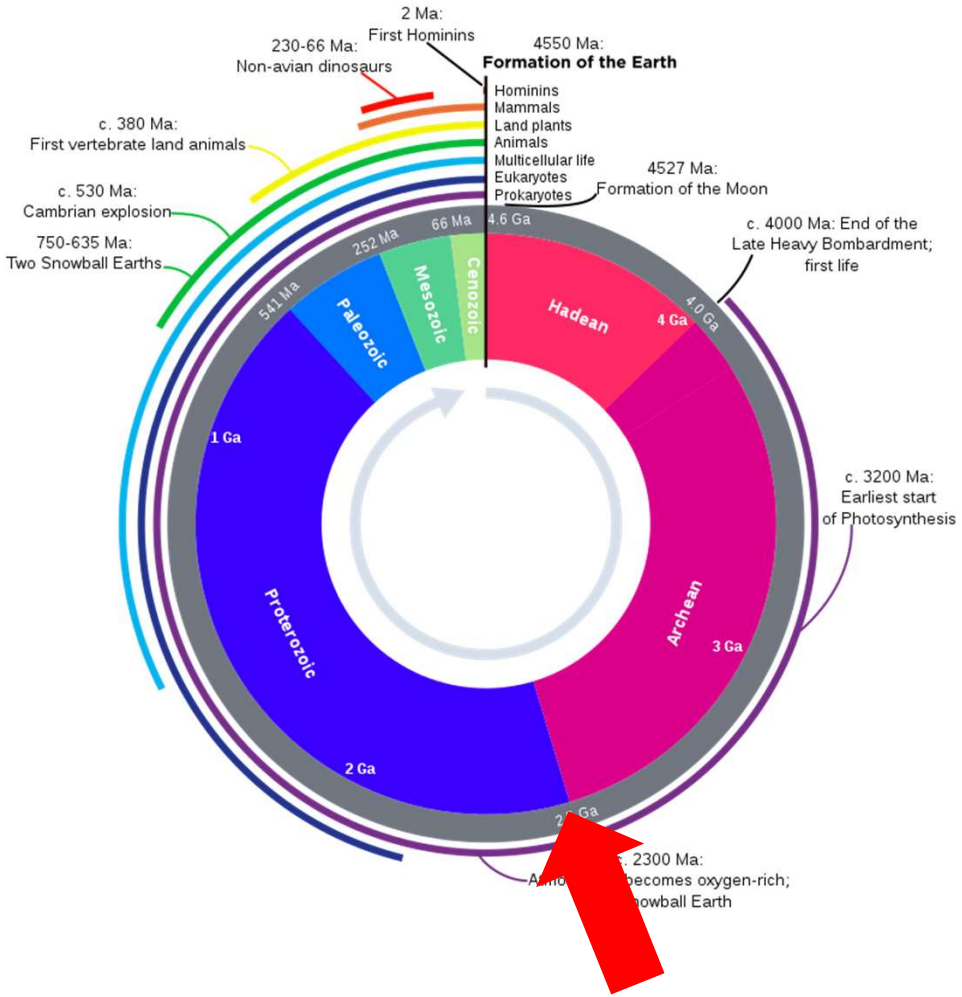
Full episode here:
https://learningonscreen.ac.uk/ondemand/index.php/prog/024F5AFE?bcast=78934002

Bands of iron oxides show evidence of oxygen production



By André Karwath, CC BY-SA 2.5, https://commons.wikimedia.org/w/index.php?curid=295589

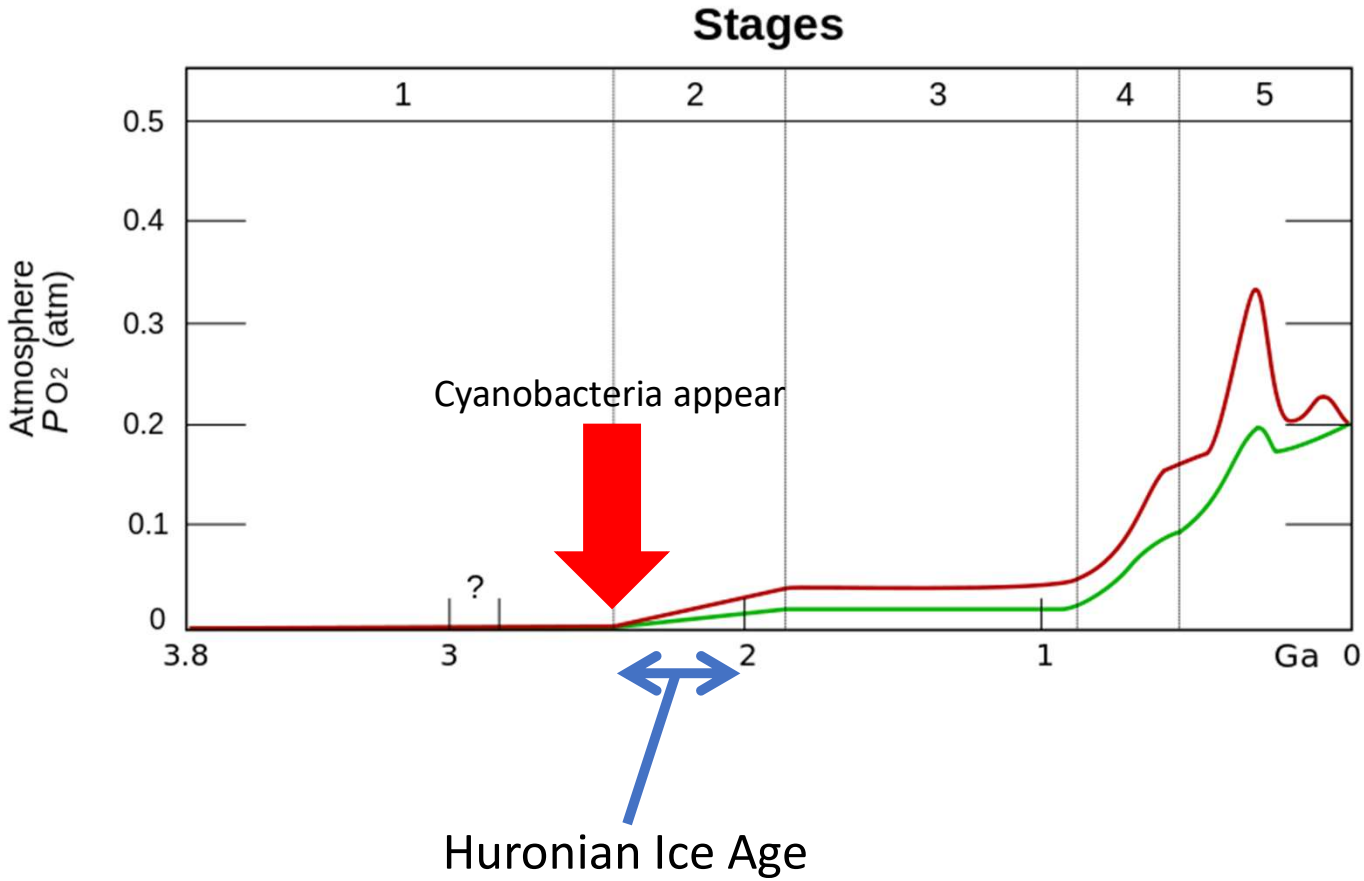
The Oxygen Crisis…



Huronian Ice Age



The Oxygen Crisis…



By Heinrich D. Hollandderivative: CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=12776502

Negatives of Oxygen

$$O_2 + Fe^{2+} \rightarrow Fe^{3+}\ O_2^{-\bullet}$$

Superoxide – a free radical (unpaired electron)

$$2\ O_2^{-\bullet} + 2H^+ \rightarrow H_2O_2 + O_2$$

Hydrogen Peroxide - bleach

$$H_2O_2 + Fe^{2+} \rightarrow OH^{\bullet} + OH^- + Fe^{3+}$$

Hydroxyl radical – VERY reactive

Fenton Reaction

For a short video about free radicals: https://www.youtube.com/watch?v=Q7AZiX6x56I

Negatives of Oxygen



All of Which Meant…



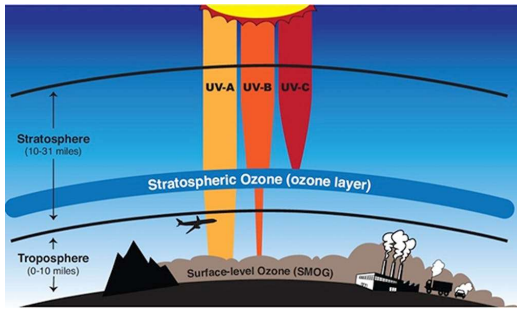
Most life on Earth disappeared…

…but not all!

Positives of Oxygen

$$O_2 + UV \rightarrow 2O^{\bullet}$$

$$O_2 + O^{\bullet} \rightarrow O_3$$



Ozone

Ozone forms a layer in the stratosphere

Prevents UV light reaching the surface

Allows life to live outside water

https://www.youtube.com/watch?v=OiwUDvd22ao

Positives of Oxygen

Without Oxygen (anaerobic respiration):

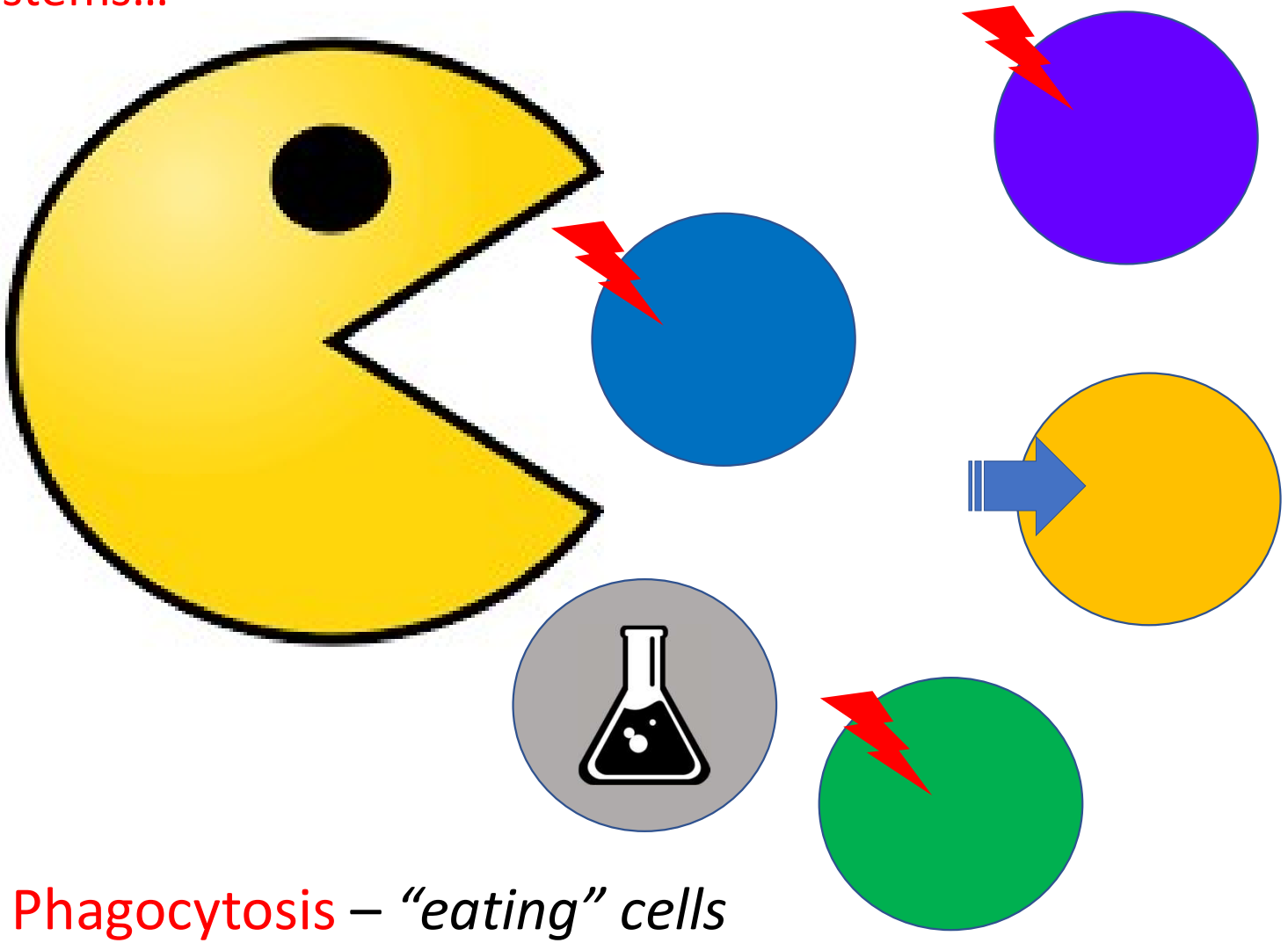
$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 2ATP$

With Oxygen (aerobic respiration)

$C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O + 36ATP$

Better respiration means BIGGER cells

Early Ecosystems…



Phagocytosis – *“eating” cells*

Interlude

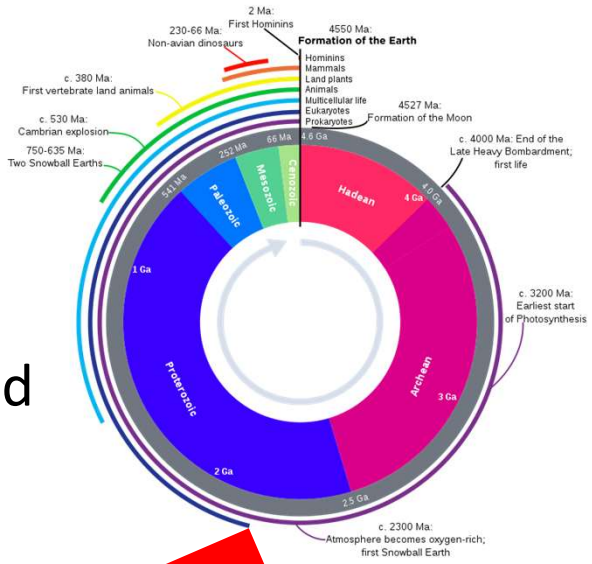
EART11100
How to Grow a Planet



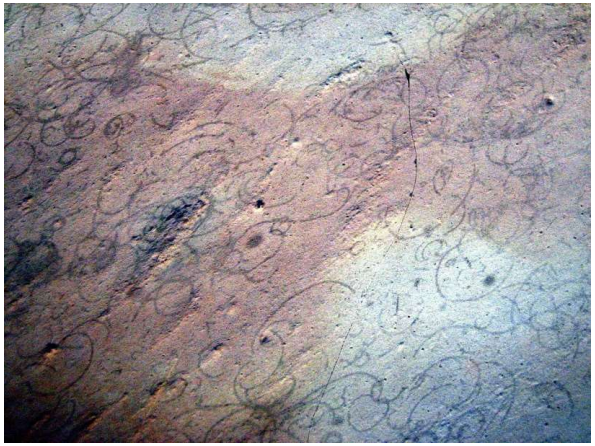
Part 2: Come Together

~2 Billion year ago…

- Life has begun
- Photosynthesis has evolved
- Cyanobacteria have made O_2
- Huronian Ice Age comes to an end



Grypania spiralis –
possibly the oldest eukaryote
c. 1.9 Billion years old

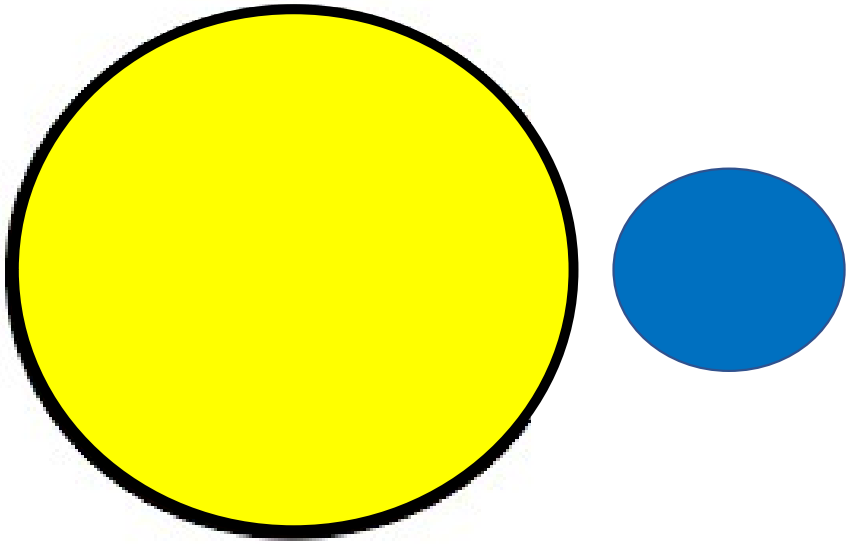


By Xvazquez, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=4348307

Complex cellular structures appear

Eukaryotes

Phagocytosis – the first heterotrophs



Aerobic respiration more efficient - allowed bigger cells to evolve

Big cells can engulf little cells

(…or little cells can invade big cells)

Requires structures for engulfing (actin cytoskeleton…)

and for digestion

What if the digestion step goes wrong?

https://en.wikipedia.org/wiki/Symbiogenesis

Endosymbiosis

Symbiosis:

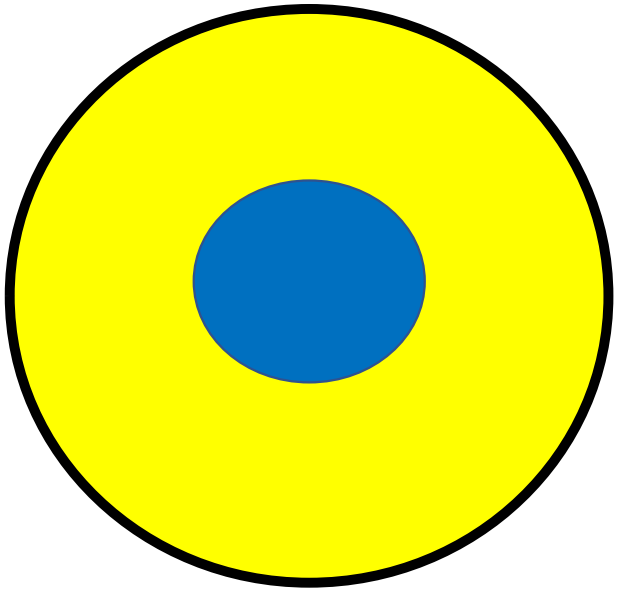
persistent living together of different organisms

Parastitic – one benefits, the other is harmed

Commensal– one benefits, the other is unaffected

Mutualistic – both benefit

Endosymbiosis – one organism lives inside another

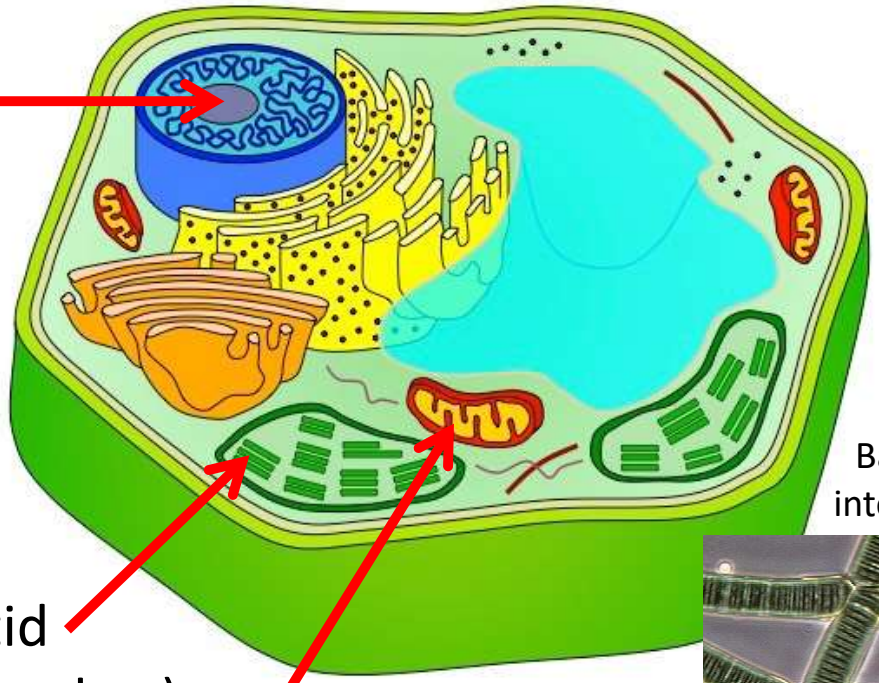




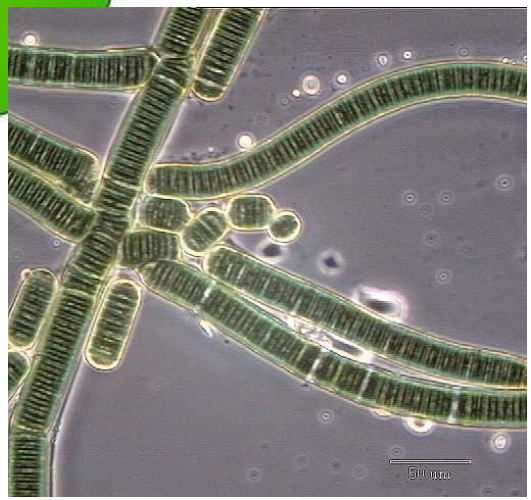
Lynn Margulis

The Eukaryotic Cell

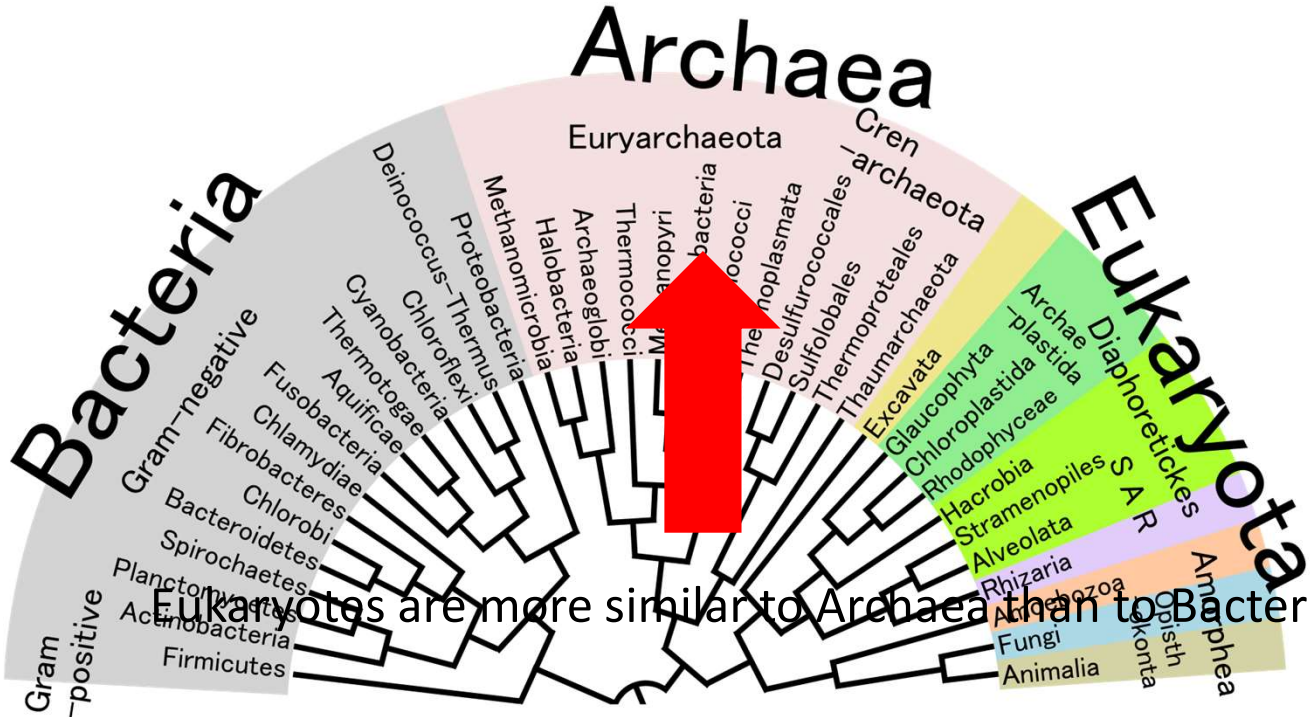
Organelles!



Bacteria can have internal membranes



Origins of Eukaryotes



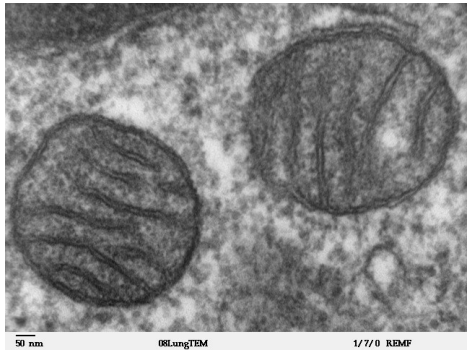
Eukaryotes are more similar to Archaea than to Bacteria

Eukaryotes probably evolved from Archaea

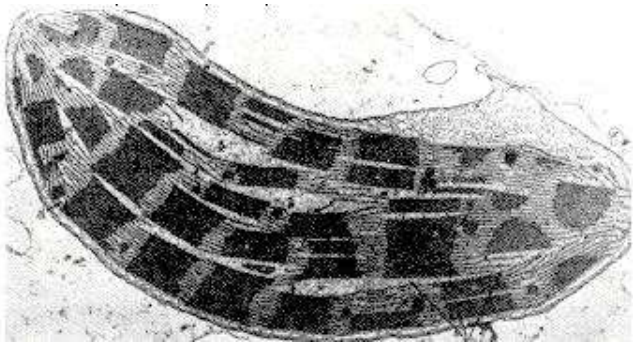
Features of Mitochondria and Chloroplasts

- Surrounded by double membrane – inner membrane controls entry/exit
- Contain DNA – simple genome, lacks essential genes
- Can make proteins
- Possess characteristics of bacteria – e.g. 70S ribosomes, codon usage

Evolved from Bacteria



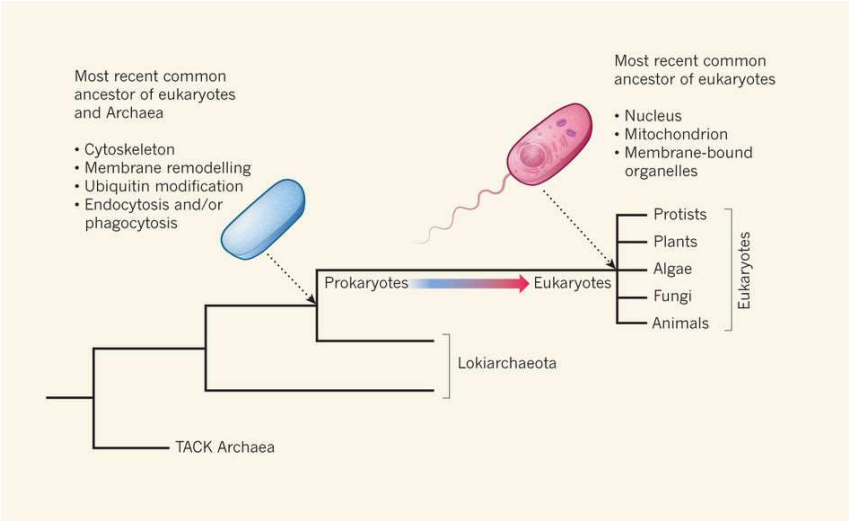
By Louisa Howard -,Public Domain, https://commons.wikimedia.org/w/index.php?curid=1248089



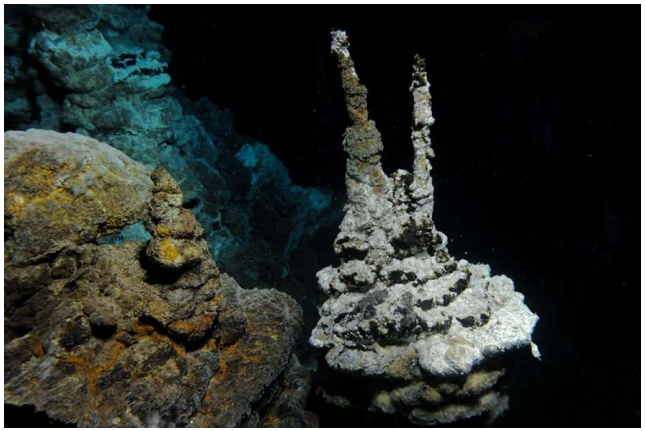
By המרכז להוראת מדעים, האוניברסיטה העברית בירושלים, GPL, https://commons.wikimedia.org/w/index.php?curid=884931

Lokiarchaeota- our closest Archaean relative?

Possesses traits of eukaryotes, *including phagocytosis*



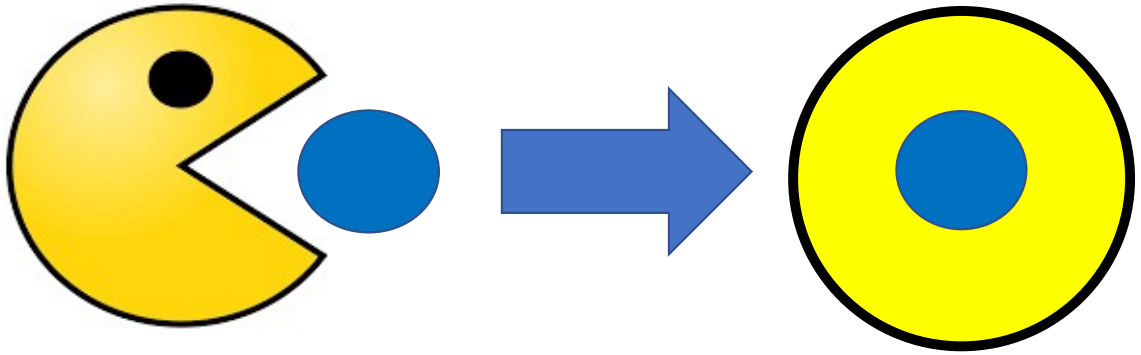
Loki's Castle, near Norway
Discovered 2008



http://www.nature.com/nature/journal/v521/n7551/full/nature14447.html

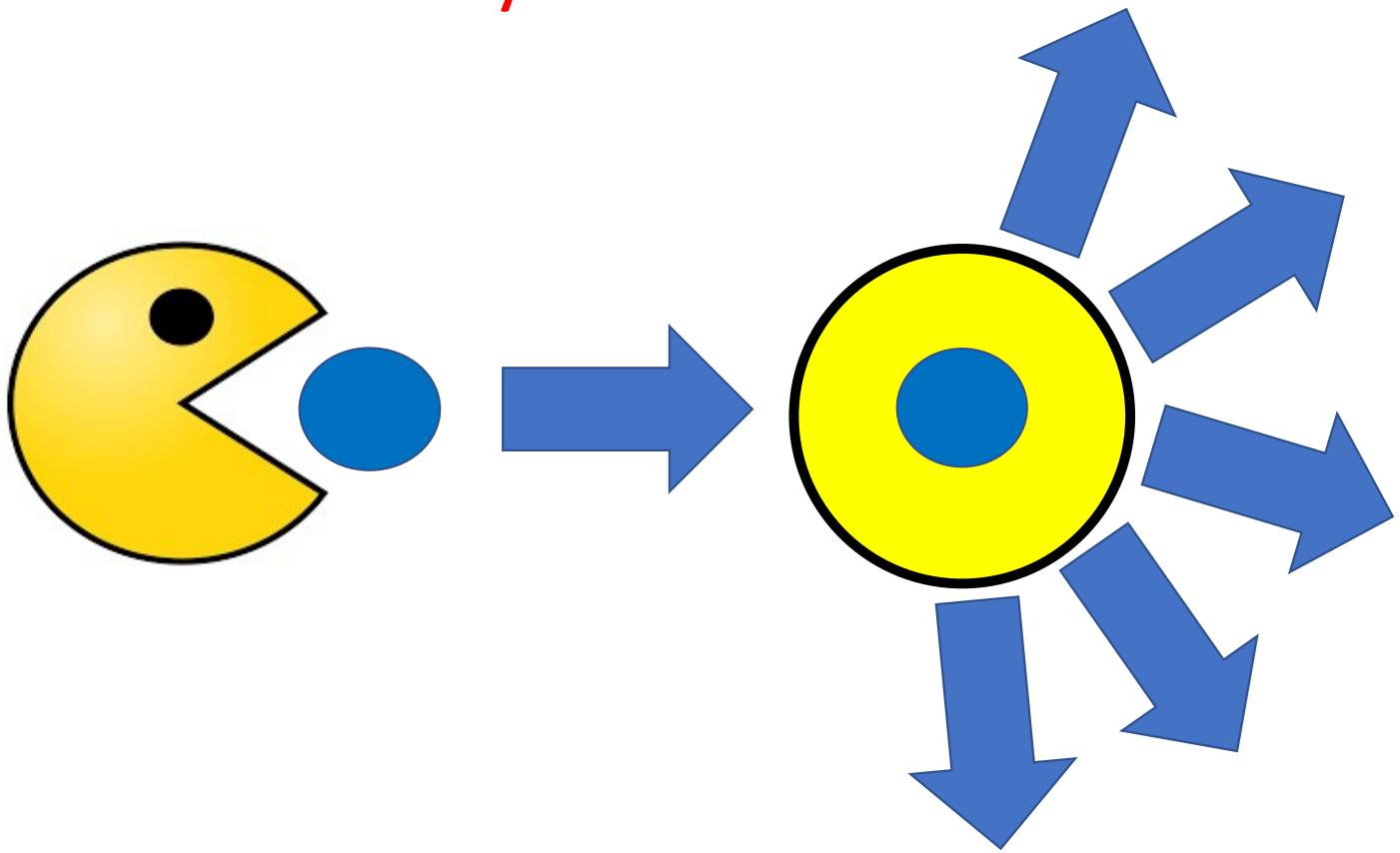
Or see: https://en.wikipedia.org/wiki/Lokiarchaeota

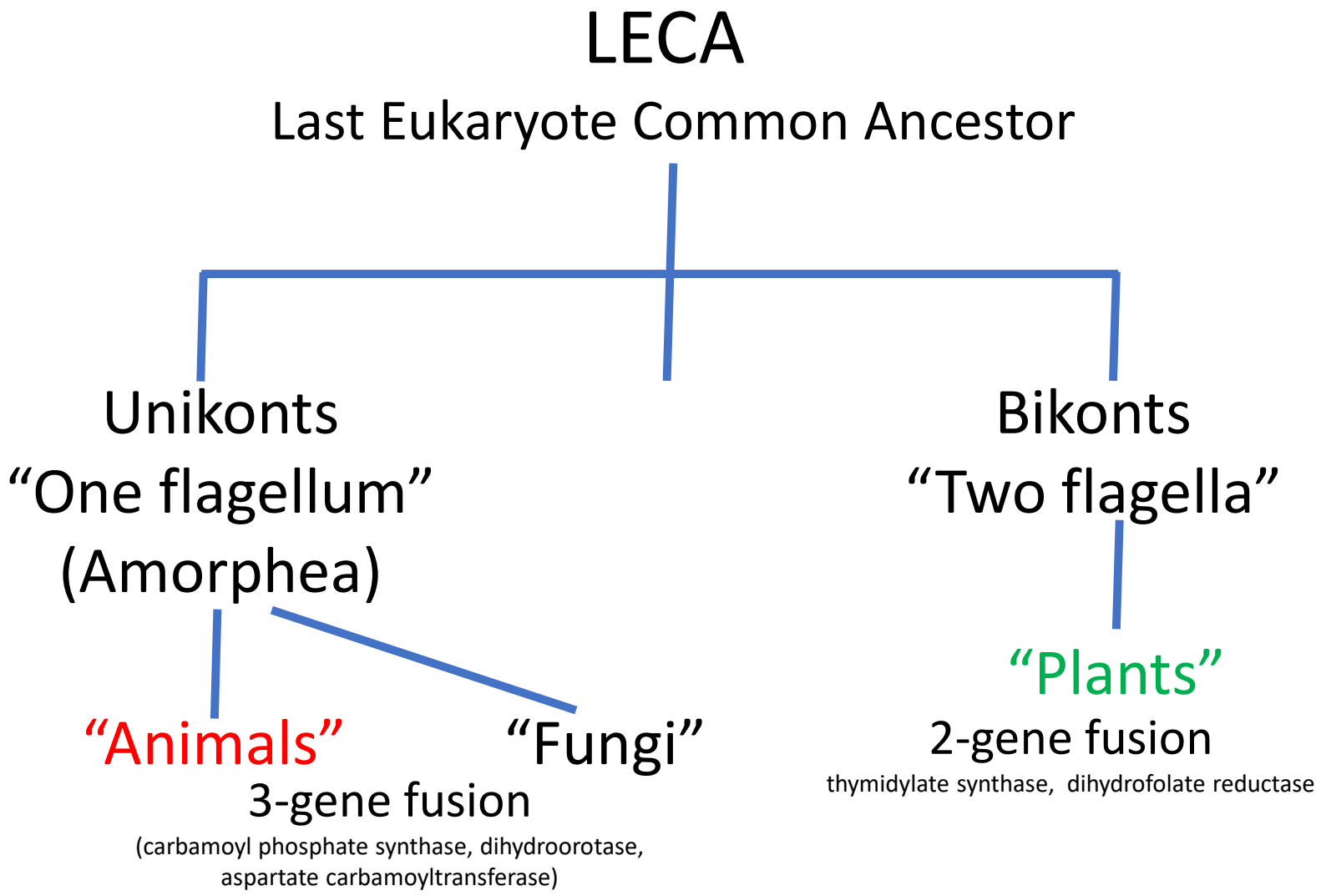
A Happy Marriage?



- Organelles have lost (some) genes
- So, “lost” control?
- Organelles “talk” to nucleus (retrograde signalling)
- And visa versa (anterograde signalling)
- Organelle “feeds” host
- But host “feeds” organelle…

Mitochondria and Chloroplasts are each thought to be the result of a single endosymbiotic event



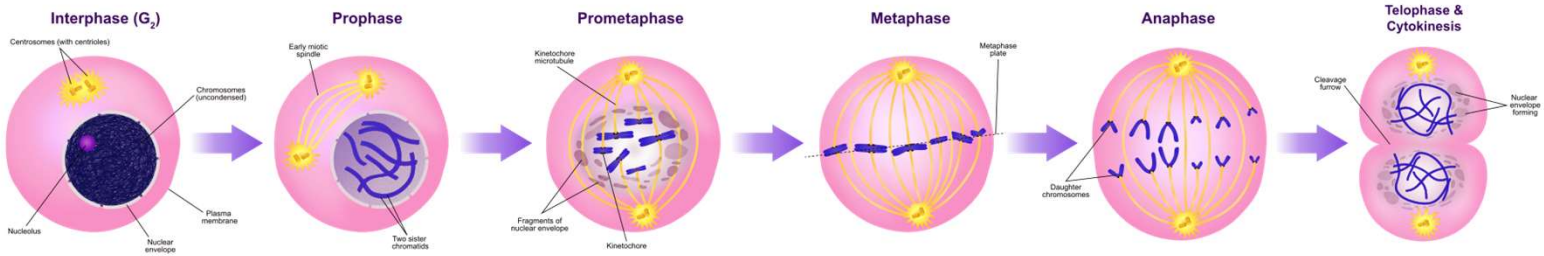

LECA
Last Eukaryote Common Ancestor
Unikonts
"One flagellum"
(Amorphea)
Bikonts
"Two flagella"
"Animals"
"Fungi"
3-gene fusion
(carbamoyl phosphate synthase, dihydroorotase,
aspartate carbamoyltransferase)
"Plants"
2-gene fusion
thymidylate synthase, dihydrofolate reductase

Divide and conquer…

Eukaryotes evolved a nucleus with DNA arranged in chromosomes

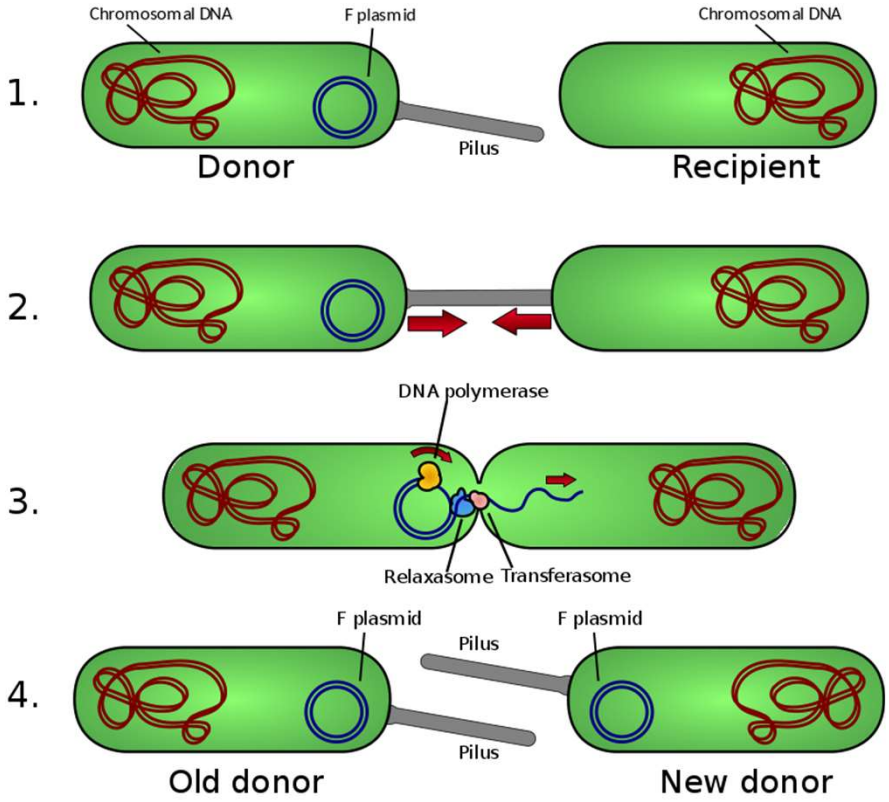
These must be shared between daughter cells

Mitosis!



By Ali Zifan - Own work; Used information from: Campbell Biology (10th Edition) by: Jane B. Reece & Steven A. Wasserman.and Nature.com., CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=49721676

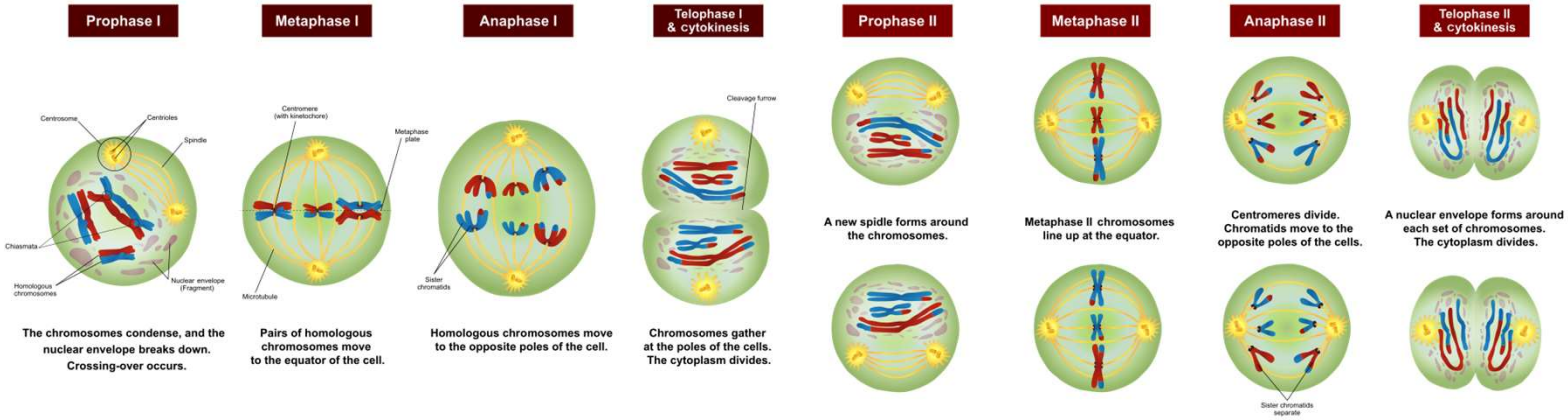
Let's talk about sex…



Bacterial conjugation – sex but not as we know it?

Let's talk about sex…

Meiosis



Genomes fuse, shuffle and divide

End with the same genes…

…but a different set of alleles

Further Reading:

(to help you understand key concepts)

Sadava Life – Chapter 5.3 – Eukaryotes

Chapter 11.3-11.5 Mitosis and Meiosis

Further watching:

(for fun – not examinable) Lecture by Nick Lane on the origins of eukaryotes

https://youtu.be/gaXhkZoOOYc

In summary…

- Oxygen allows for bigger cells – more efficient respiration
- Endosymbiosis – eukaryotes evolve more complex cell structures with shared bacterial and archaeal features
- A single endosymbiotic event leads to each organelle type
- Sexual reproduction allows for efficient "shuffling" of genes

Next week: Life Gets Large and Complicated…
…with Rob Sansom